



BUSINESS ANALYSIS FOR SOFTWARE PROJECTS

4. Requirements Elicitation

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4.1 What is a Requirement

- A requirement is some aspect of a system's content or behavior, which is necessary or desired by the customer or client
 - Provides a clear understanding of the problem domain
 - Establishes a context of the need/problem
- Requirements define
 - *2- What* the system must do
 - *What* characteristics or properties the system must have
 - *3- What* information is involved
 - *What* degree of quality is expected
 - *What* constraints apply to the solution
 - *1- Who* will use the system
 - *4- How* the system is to be used

4.1 What is a Requirement

- It may range from a **high-level abstract** statement of a service or of a system constraint to a **detailed mathematical** functional specification
- This is inevitable as requirements may serve a dual function
 - May be the basis **for a bid** for a contract - therefore must be **open** to interpretation
 - May be the basis for the **contract itself** - therefore must be defined in **detail**.
 - Both these statements may be called requirements.

4.1 What is a Requirement

- **Well defined requirements are**
 - **Needed – Discrete**
 - **Correct – Unique**
 - **Complete – Verifiable**
 - **Concise – Traceable**
 - **Unambiguous – Consistent**

4.2 Requirement Characteristics

It is important to recognize that needs, wants, and expectations are not requirements until they have been elicited, analyzed, and documented

The desired characteristics of a requirement include the following:

- *Solution free or technology independent.*
- *Organized.* A hierarchical structure is preferred. The structure should ensure that the requirements are complete, readily locatable, sorted by type, and integrated without conflicting with each other.
- *Clear.* Short, simple, unambiguous requirement statements are preferable; however, any qualifications regarding the requirements should be noted.
- *Prioritized.* The value of each stakeholder requirement may be ranked.
- *Traceable.* Each requirement should be specific to a stakeholder or other origin. This facilitates the change management process

4.2 Requirement Types

- **Functional requirements**

Describe the interactions between the system and its environment independently from the implementation

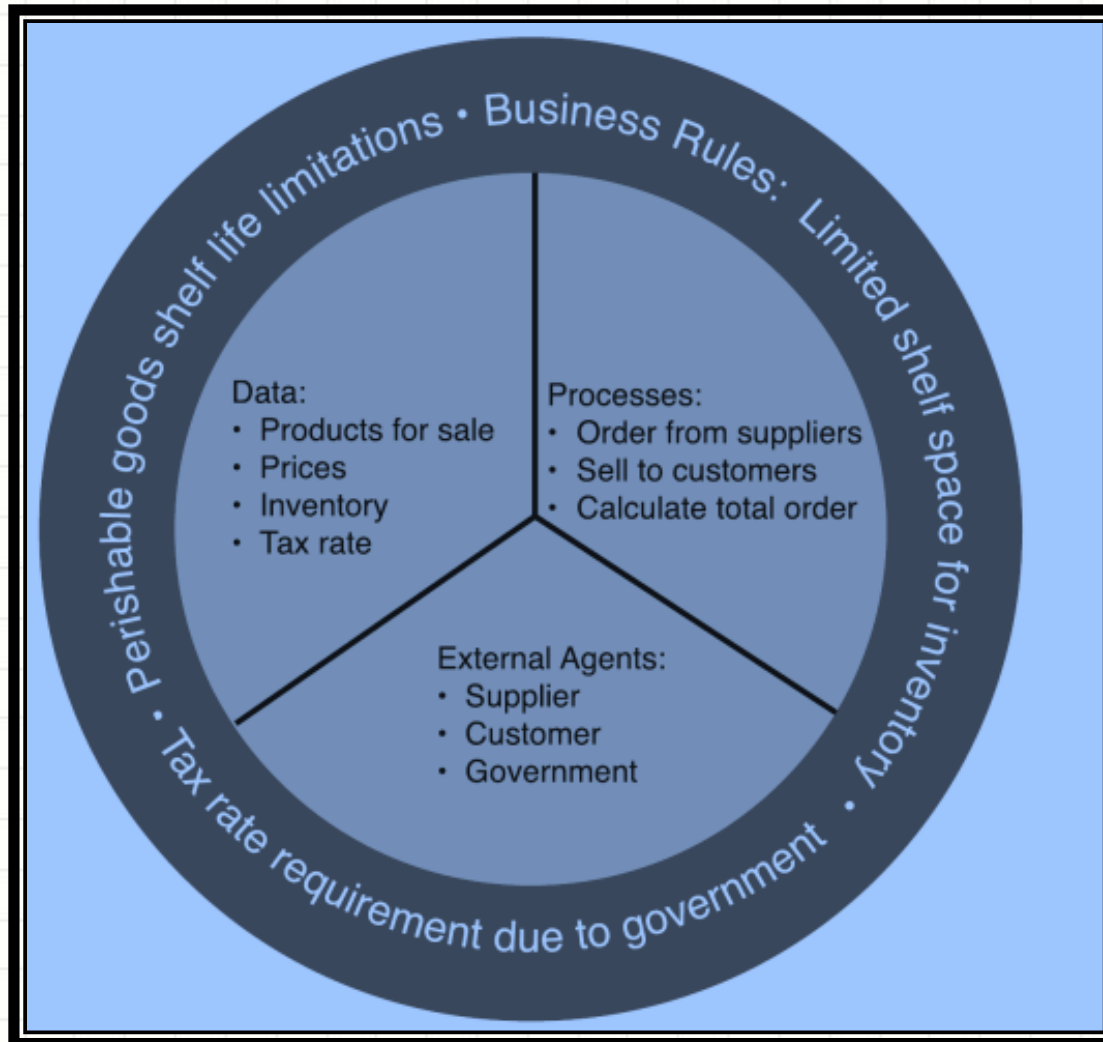
- **Nonfunctional requirements**

Aspects not directly related to functional behavior

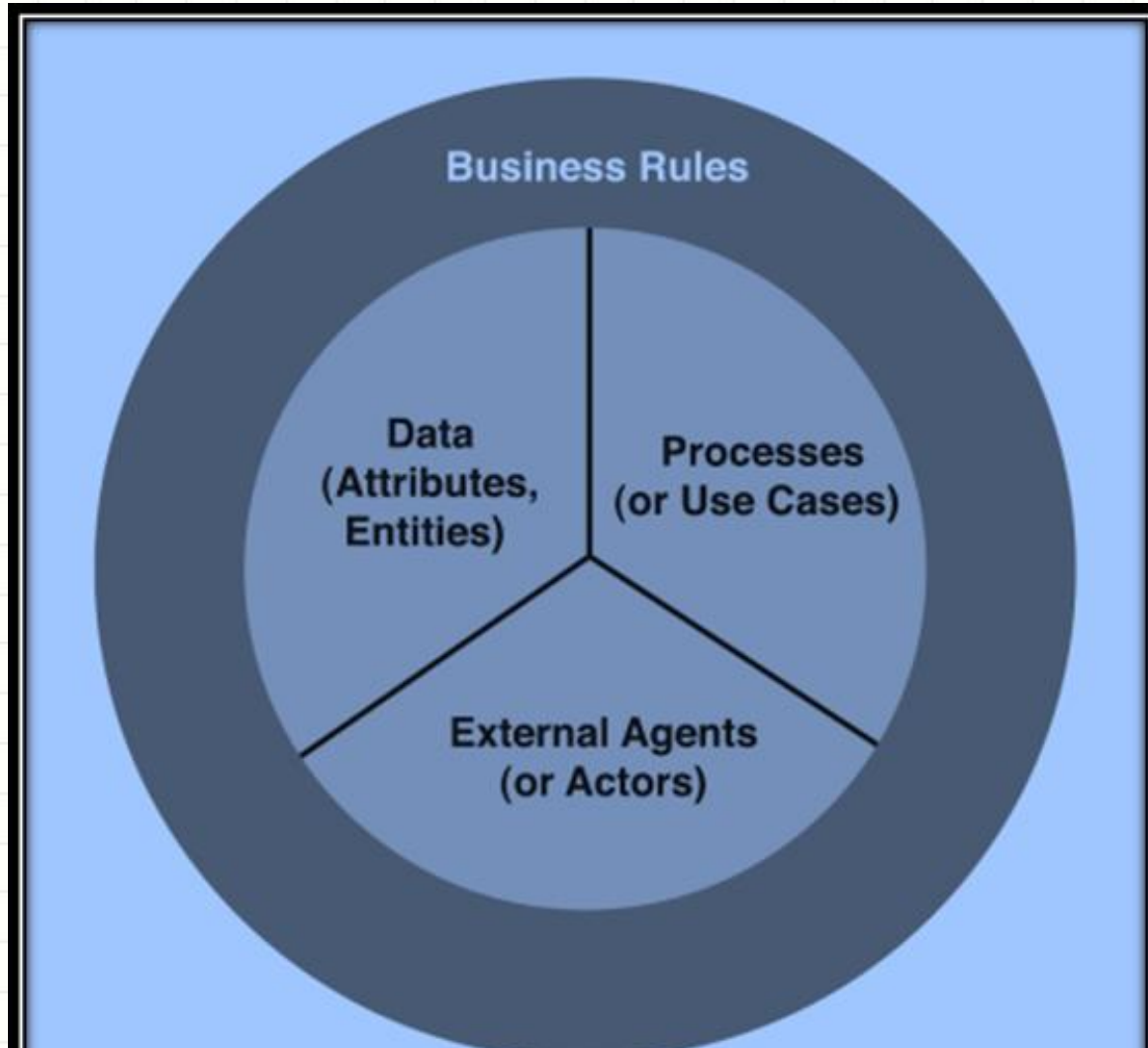
“The response time must be less than 1 second”

- **Business Rules**

4.2 Requirement Types



4.2 Requirement Types



4.3 Requirements Attributes

Complexity is not mentioned in the requirements documents

<u>Complexity</u>	• A judgment of how hard a requirement will be to implement.
<u>Absolute Reference</u>	• Unique identifier for the requirement. • Not to be altered or re-used if the requirement is moved, changed or deleted.
<u>Risks</u>	• Risk associated with meeting or not meeting the requirement.
<u>Author</u>	• Who documented the requirement. • If the requirement is later found to be ambiguous the author may be consulted for clarification.
<u>Source</u>	• Of who provided the requirement. • Every requirement needs a source with authority to specify and approve requirements.

4.3 Requirements Attributes

<u>Status</u>	• Current status of the requirement. • Indicates things such as whether it is proposed, accepted, verified with the users, or implemented.
<u>Ownership</u>	• Individual or group needing the requirement or will be the business owner after the product is released.
<u>Urgency</u>	• Specifies when the requirement is needed. • Only specify separately from “Priority” when an implementation deadline exists.
<u>Priority</u>	• Which requirements to implement first based on importance to the stakeholders.
<u>Stability</u>	• Indicates how mature the requirement is. • Used to determine if a requirement is firm enough to start work on.

Why Define Requirements

- Poor-defined requirements and unmanaged requirement changes lead to
 - **Disagreement** with customers on requirements and product acceptance
 - **Unrealistic** estimates and commitments to customers
 - Schedule **slippage**
 - Cost **overruns**
 - Uncertainty regarding project **closure**
 - Team **burnout**
 - **Unhappy** customers
- It is critical to involve the customers and stakeholders early in the requirements definition process

4.4 Requirements Elicitation Definitions

Requirement elicitation is just the process of **extracting** the information from users, customers, and group of people.

It is **non-trivial** as we can never find the requirement of the system from the user and customer by just asking them what the system actually needs (because customers doesn't know what they really need).

Possible **ways** to get the requirements are making questionnaires, brain storming, work-shopping and interviewing.

4.4 Requirements Elicitation Definitions

Is the practice of collecting the requirements of a system from users, customers and other stakeholders. The practice is also sometimes referred to as requirements gathering.

Requirements Elicitation focuses on understanding what is really needed to allow the business objectives to be met by the sponsor of customer.

Requirements Elicitation is formally a process of drawing out responses from various stakeholders at the client organization about the project requirements.

Example

Idea: A Library Management System should be designed. Information on books, CDs, DVDs, Journals, etc. can be stored and retrieved

Problem Statement

Possible Requirements:

- ◆ Searching by Title, Author, and/or ISBN should be possible
- ◆ User Interface should be web-based (accessible via WWW Browser)
- ◆ At least 20 transactions per seconds should be possible
- ◆ All services should be available within 10 minutes
- ◆ Users have no access to personal data of other users

functional
requirement

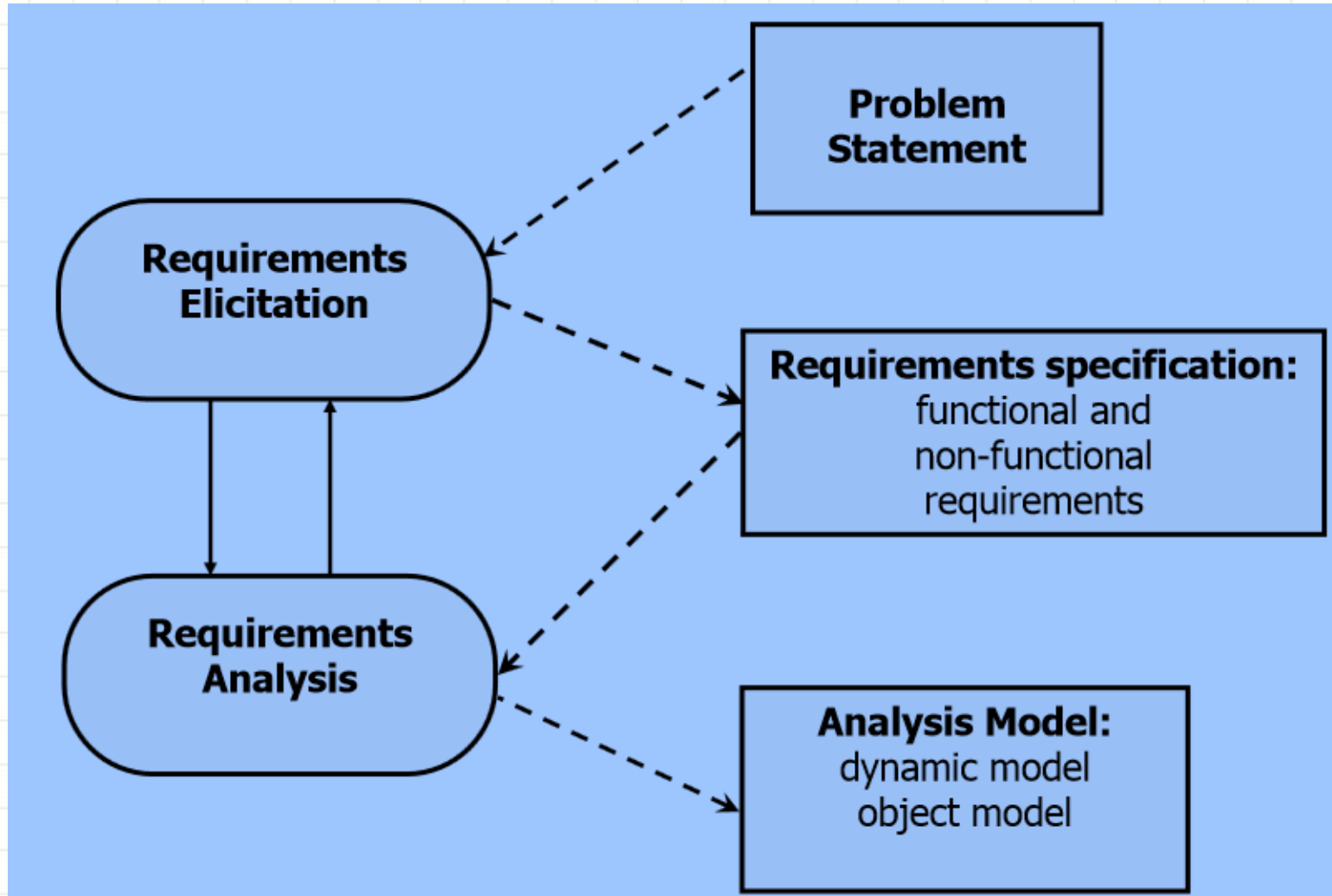
Implementation
requirement

performance
requirement

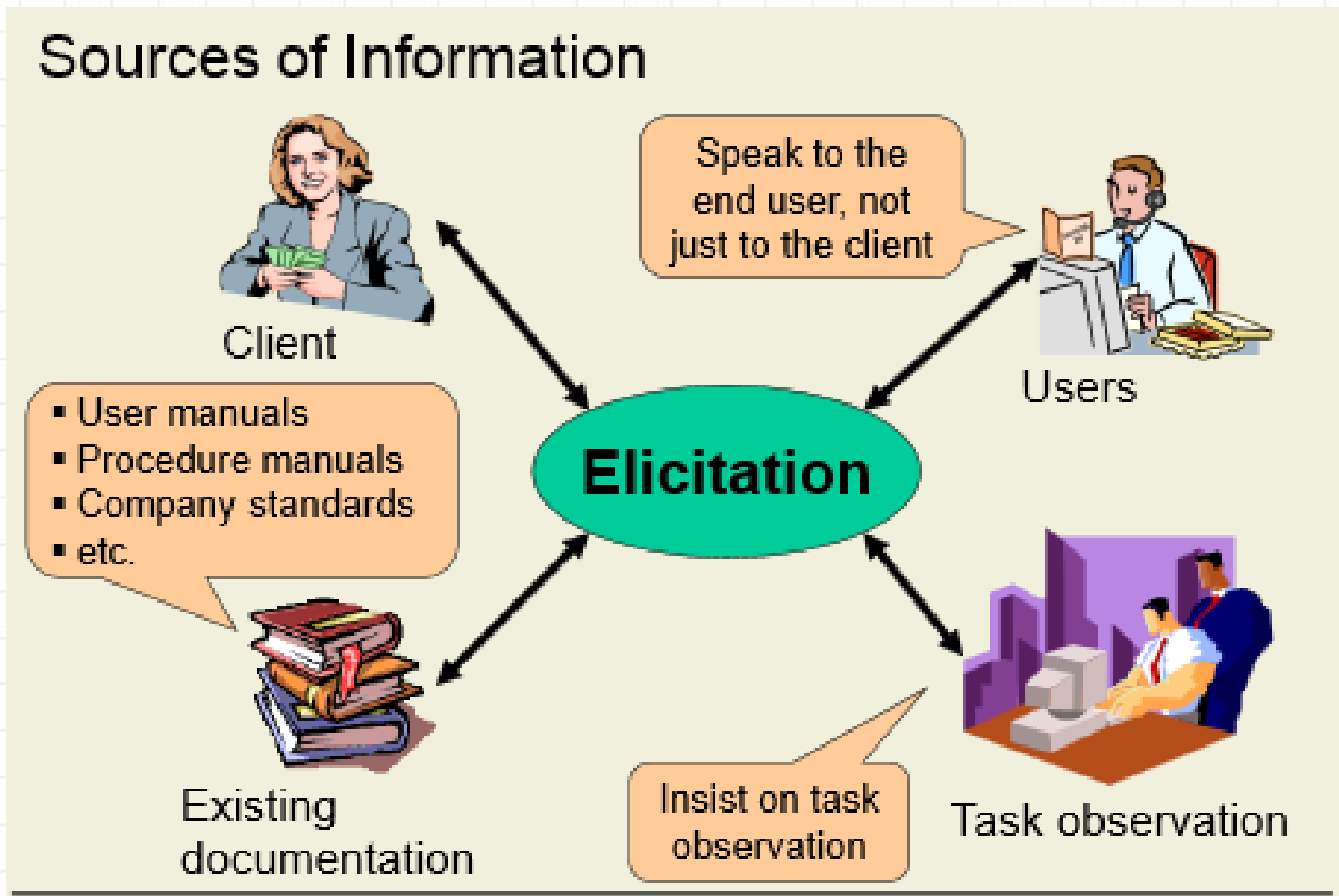
availability
requirement

Security
requirement

Requirements Elicitation Process



4.5.3 Sources of Requirements



4.5.3 Sources of Requirements

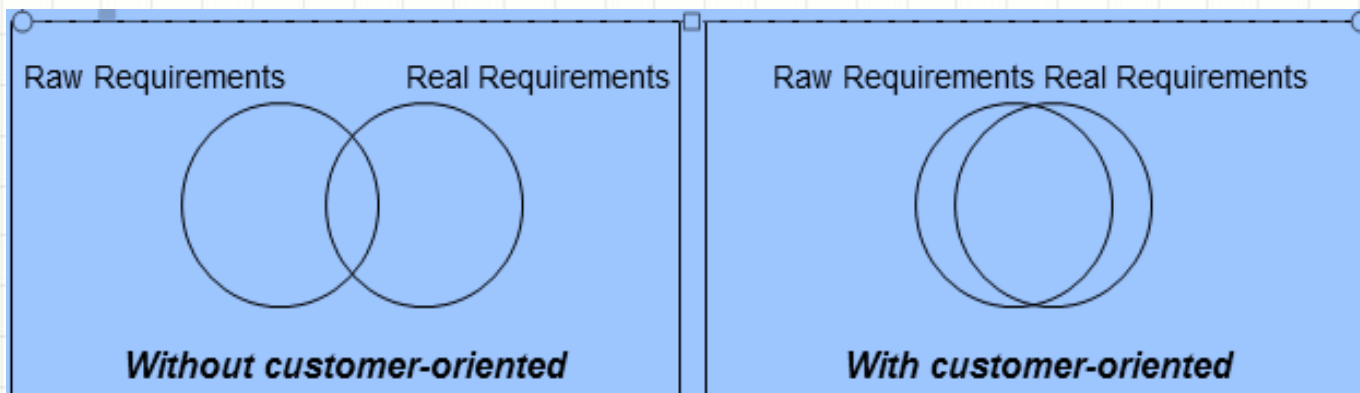
1- Stakeholders

- Stakeholders include anyone with an interest in the system, for example
 - Sponsors
 - Customers
 - System Users/Domain Experts
 - Project Team
- Identify stakeholders by asking questions
 - Who will use the product?
 - Who will provide the inputs?
 - Who will get the outputs?
 - Who has an oversight role?
 - Who has a related role?
 - Who will be rewarded?
 - Who will be penalized?

4.5.3 Sources of Requirements

1- Stakeholders

- Involving stakeholders in individual or group requirements sessions to define system details
 - Results in a **high quality system** from the customer/user point of view
 - Results in higher end-user **satisfaction**
 - Helps to **educate** customer/user about system through their **participation**



4.5.3 Sources of Requirements

1- Stakeholders

- **Benefits** of stakeholder involvement include
 - Improved system information content
 - Improved system information quality
 - Improved system productivity
 - Improved customer/user understanding of the system
 - Creation of customer/user consensus
 - Creation of customer/user commitment to system success
 - Determination of early scoping decisions
 - Increased confidence that consensus will be reached
 - Improved quality of requirements
 - Development of an integrated team

4.5.3 Sources of Requirements

4- Existing Systems

Analysis of existing **automated** systems provides insight into data that might be used by the new system.

Documents to be reviewed:

- **User Manuals**
- **Data Samples**
- **Interface Descriptions**
- **Sample Reports**
- **Sample Screens**